

CEH

(Certified Ethical Hacker)

Module 02

Footprinting and Reconnaissance

Lab Report: Certified Ethical Hacking v13 - Module 02 Task: Advanced Footprinting and Reconnaissance

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Task Objective: Perform passive and active footprinting on a given target domain “https://digiskills.pk” to gather as much publicly available information as possible without triggering intrusion detection mechanisms. Document the process, tools, and findings while emphasizing legal and ethical boundaries.

Target: Use the domain or IP address (https://digiskills.pk). **Do NOT attack or probe unauthorized live sites.**

Step 01: **Passive Reconnaissance**

Initial Reconnaissance: We use **Whois** command performed **5.9.80.16** to gather publicly available information.

Target Identification: This **5.9.80.16** <Target Ip> is part of the **HETZNER-fsn1-dc7** netname, which belongs to Hetzner Online GmbH
Address: A German hosting provider Phone no: **+49 9831 505-0**

Vulnerability/Risk: This provides a Public Abuse Contact “abuse@hetzner.com” and contact details for the organization, which could be used for Social engineering Attacks.

```
(subhan@subhan)-[~]
$ whois 5.9.80.16

% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html

% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.

% Information related to '5.9.80.0 - 5.9.80.31'

% Abuse contact for '5.9.80.0 - 5.9.80.31' is 'abuse@hetzner.com'

inetnum:        5.9.80.0 - 5.9.80.31
netname:        HETZNER-fsn1-dc7
descr:         Hetzner Online GmbH
descr:         Datacenter fsn1-dc7
country:        DE
admin-c:        HOAC1-RIPE
tech-c:         HOAC1-RIPE
status:         ASSIGNED PA
remarks:        INFRA-AW
mnt-by:         HOS-GUN
mnt-lower:      HOS-GUN
mnt-routes:     HOS-GUN
created:        2012-10-18T12:05:53Z
last-modified:  2018-03-15T14:28:56Z
source:         RIPE

role:           Hetzner Online GmbH - Contact Role
address:        Hetzner Online GmbH
address:        Industriestrasse 25
address:        D-91710 Gunzenhausen
address:        Germany
phone:          +49 9831 505-0
fax-no:         +49 9831 505-3
```

Information Gathering: This Data is Public and Easy accessible from a RIPE Database, a public database containing details for Ip Address **5.9.0.0/16** and AS numbers **AS24940** both belongs to Hetzner Online GmbH.

Identified Information: The query provides administrative information a peering mailbox “peering@hetzner.com” which are useful for security research to vulnerabilities

No Active Vulnerabilities: This information does not reveal any active vulnerabilities.

```
fax-no: +49 9831 505-3
abuse-mailbox: abuse@hetzner.com
remarks: *****
remarks: * For spam/abuse/security issues please contact *
remarks: * abuse@hetzner.com, or fill out the form at *
remarks: * abuse.hetzner.com, thank you. *
remarks: *****
remarks: *****
remarks: * Any questions on Peering please send to *
remarks: * peering@hetzner.com *
remarks: *****
org: ORG-HOA1-RIPE
admin-c: MH375-RIPE
tech-c: GM834-RIPE
tech-c: SK2374-RIPE
tech-c: MF1400-RIPE
tech-c: SK8441-RIPE
tech-c: DD15478-RIPE
nic-hdl: HOAC1-RIPE
mnt-by: HOS-GUN
created: 2004-08-12T09:40:20Z
last-modified: 2022-11-22T18:33:55Z
source: RIPE # Filtered

% Information related to '5.9.0.0/16AS24940'

route: 5.9.0.0/16
descr: HETZNER-RZ-FKS-BLK5
origin: AS24940
mnt-by: HOS-GUN
created: 2012-04-26T10:30:12Z
last-modified: 2012-04-26T10:30:12Z
source: RIPE

% This query was served by the RIPE Database Query Service version 1.118.1 (BUSA)
```

01: The **dig** command was used for passive footprinting to perform a DNS lookup on the target domain digiskills.pk.

02: The query successfully returned the domain's **IPv4** address (**A Record**), which is **58.27.207.132**.

03: The DNS query was sent from the local machine to the server at **192.168.100.1**, which is likely a local router.

The **nslookup** command was executed to perform DNS recon on the domain digiskills.pk. The query revealed that the domain resolves to the IPv4 address **5.9.80.16**. This information is considered a public finding and is a fundamental first step in the reconnaissance phase of a penetration test. The IP address ownership can be investigated further to identify the hosting provider or server location, This does not contain any Vulnerability.

```
(subhan@subhan)-[~]
$ dig digskills.pk

; <<>> DiG 9.20.7-1-Debian <<>> digskills.pk
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 33713
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;digskills.pk.                IN      A

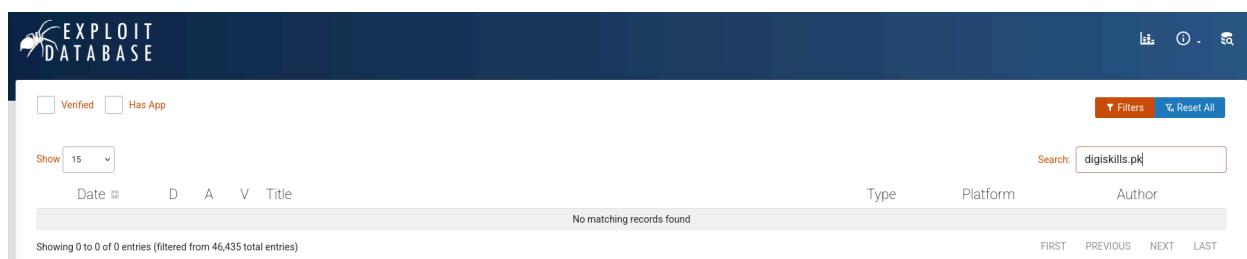
;; ANSWER SECTION:
digskills.pk.                253     IN      A      58.27.207.132

;; Query time: 3 msec
;; SERVER: 192.168.100.1#53(192.168.100.1) (UDP)
;; WHEN: Fri Jul 04 12:24:22 PKT 2025
;; MSG SIZE rcvd: 47
```

```
(subhan@subhan)-[~]
$ sudo nslookup digskill.pk
[sudo] password for subhan:
Server:                192.168.0.1
Address:                192.168.0.1#53

Non-authoritative answer:
Name:   digskill.pk
Address: 5.9.80.16
```

Google Dorking For related info:Based on a Search of the Exploit Database,no publicly documented Info was found for the platform “[digskills.pk](#)” at the time of this report. The search was performed against a database containing over 46,000 total entries, but the results were filtered to show zero matching records.



01: The Subdomains “lms.digiskills.pk” and “mail.digiskills.pk” are hosted on the same IP address, **58.27.207.132**, and are protected by Cloudflare.

02: The Subdomains “www.digiskills.pk” is hosted on a different IP address, **111.68.104.221**, and does not appear to be using Cloudflare.

Subdomain	IP	Cloudflare
lms.digiskills.pk	58.27.207.132	
mail.digiskills.pk	58.27.207.132	
www.digiskills.pk	111.68.104.221	
IP		Count
58.27.207.132		2
111.68.104.221		1

Host Identification: The host at IP address **58.27.207.132** is located in Lahore, Pakistan, and is associated with the organization “**National Wimax/IMS environment**” and the domain “digiskills.pk”.

Operating System: The operating system is identified as Windows.

Web Server: The Web Server in use is Microsoft **IIS httpd10.0**.

Open Ports and Services: The only open port identified is **443/TCP**, which is used for **HTTPS** traffic.

SSL Certificate: The **SSL** certificate is issued by sectigo Limited for the domain “digiskills.pk” .

Potential Vulnerabilities: Microsoft **IIS 10.0**, like any web server software, may have known vulnerabilities if not regularly updated or properly configured. Past vulnerabilities in **IIS** have included issues like cross-site scripting (**XSS**) and elevation of Privilege (**CVE-2017-0055**).

Recommendations: It is highly recommended to ensure the server is fully patched with the latest security updates from Microsoft to mitigate known vulnerabilities.

Regular vulnerability scanning and penetration testing should be performed to identify and address potential security flaws.

A technical report on the IP address **58.27.207.132** indicates that it is Microsoft Windows Server located in Lahore, Pakistan, operated by **WATEEN-IMS-PK-AS-AP**. The server has multiple services running, including **HTTP** on port **80** and **HTTPS** on port **443**. The **HTTP** service is configured with a default title of “**IIS Windows Server**”, which could be a misconfiguration or a potential vulnerability, as it may reveal information about the server’s identity to attackers. The use of unencrypted **HTTP** on port **80** poses a security risk, as traffic is not encrypted and is vulnerable to man-in-the-middle attack and data exposure.

The screenshot displays a network intelligence tool interface. At the top, a search bar contains the query "host:ip = '58.27.207.132'". Below the search bar, the IP address "58.27.207.132" is highlighted, with a "VPN" icon and a "HOST" label. The interface is divided into several sections: "Summary", "Network and Geolocation", and "Services".

The "Summary" section shows the following details:

- Reverse DNS: mail.digiskills.pk
- Forward DNS: www.digiskills.pk, iromail.vu.edu.pk, mail.digiskills.pk
- OS: Microsoft Windows
- 4 Total Services: 80 / HTTP, 443 / HTTP, 500 / IKE, 4430 / HTTP

The "Network and Geolocation" section provides the following information:

- WHOIS Network: National Wimax/IMS environment
- WATEEN-TEL
- WHOIS Organization: No Data
- Autonomous System: 58 27.207.0/24 via WATEEN-IMS-PK-AS-AP National WIMAXIMS environment, PK (AS38264)

A map shows the location of the IP address in Lahore, Pakistan (PK), with coordinates 31.558° N, 74.35071° E.

The "Services" section is currently selected, showing details for "HTTP 80 /TCP". The "LAST OBSERVED" date is JUL 03, 2025, at 10:53 UTC. The "DETAILS" section shows the following information:

- URI: http://58.27.207.132/
- Status: 200 OK
- Path: /
- Body Hash: sha256:557b5e9015b4963a130c7039e077b3ec0306d407b6ee13d631721d5d969d2917
- HTML Title: IIS Windows Server
- Response Body: <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"> <html xmlns="http://www.w3.org/1999/xhtml"> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" /> <title>IIS Windows Server</title> <style type="text/css"> <!-- body { color:#000000; background-color:#0072C6;... more >

01: The system is running **IIS Windows Server**, which may have known vulnerabilities depending on the specific version and patch level.

02: The Transport Layer Security (**TLS**) handshake is listed as **UNKNOWN**, which could indicate a misconfiguration, an outdated protocol, or a potential issue with a certificate chain.

03: The server is using a certificate issued to “digiskills.pk” from **sectigo limited**, but the **TLS** handshake status of “**UNKNOWN**” suggests that the certificate may not be fully trusted.

HTTP 443 /TCP

LAST OBSERVED JUL 04, 2025 | 06:51 UTC

DetailsRaw DataJSON

DETAILS

URI

https://58.27.207.132/Go ↗

Status

200 OK

Path

/

Body Hash

sha256:557b5e9015b4963a130c7039e077b3ec0306d401b6ee13d631721d5d969d2917

HTML Title

IIS Windows Server

Response Body

<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"> <html xmlns="http://www.w3.org/1999/xhtml"> <head> <meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1" /> <title>IIS Windows Server</title> <style type="text/css"> <!-- body { color:#000000; background-color:#0072C6;--> </style> </head> <body> </body> </html>

TLS HANDSHAKE

Version Selected

UNKNOWN

CERTIFICATE

Fingerprint

8637c86ef2ddae2f0690752da168a010fefb32e1c54a2a7227253f788e282d96

Subject DN

CN=*.digiskills.pk

Issuer DN

C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication CA DV R36

Names

*.digiskills.pk, digiskills.pk

IKE 500 /UDP

VPN

LAST OBSERVED JUL 03, 2025 | 20:34 UTC

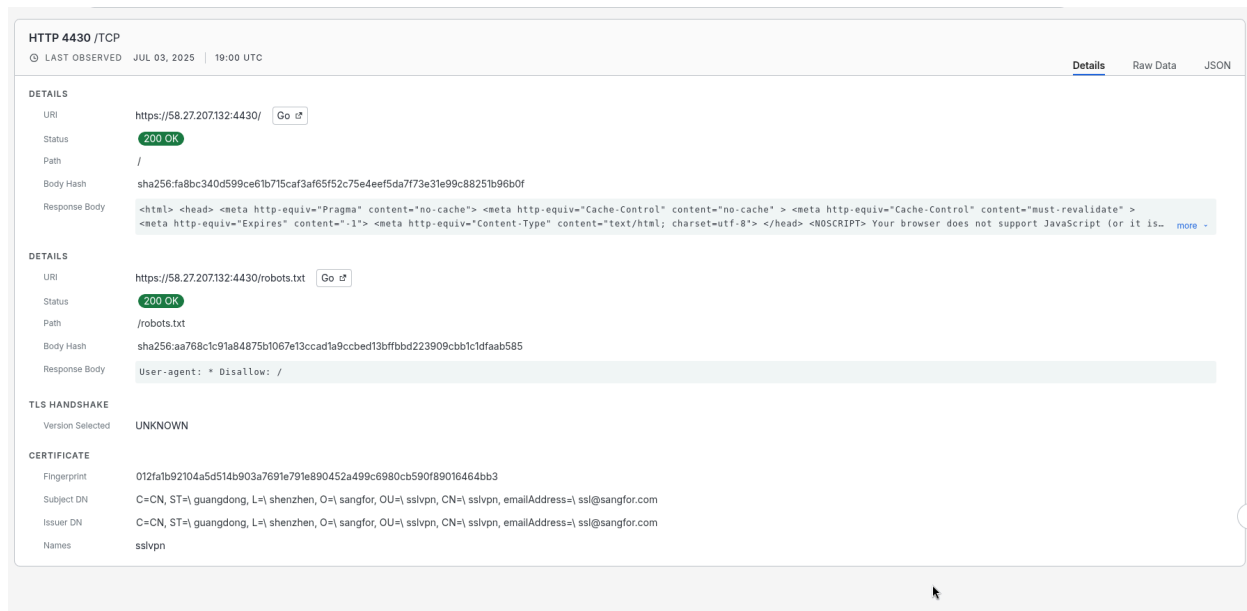
DetailsRaw DataJSON

DETAILS

Vulnerability: An Unencrypted **HTTP** service is running on **port 4430** of IP address **58.27.207.132**, which appears to be a Sangfor **SSL** VPN appliance. Although the service responds with an html page, the **TLS** handshake status is "**UNKNOWN**", indicating a lack of secure communication for the service itself.

Impact: A Potential attacker could exploit this misconfiguration to intercept sensitive data, including authentication Credentials, and potentially gain unauthorized access.

Recommendations: It is recommended to secure all internet-facing services with proper **TLS** encryption. All services should be configured to use **HTTPS** and any Unnecessary ports should be closed to reduce the attack surface.

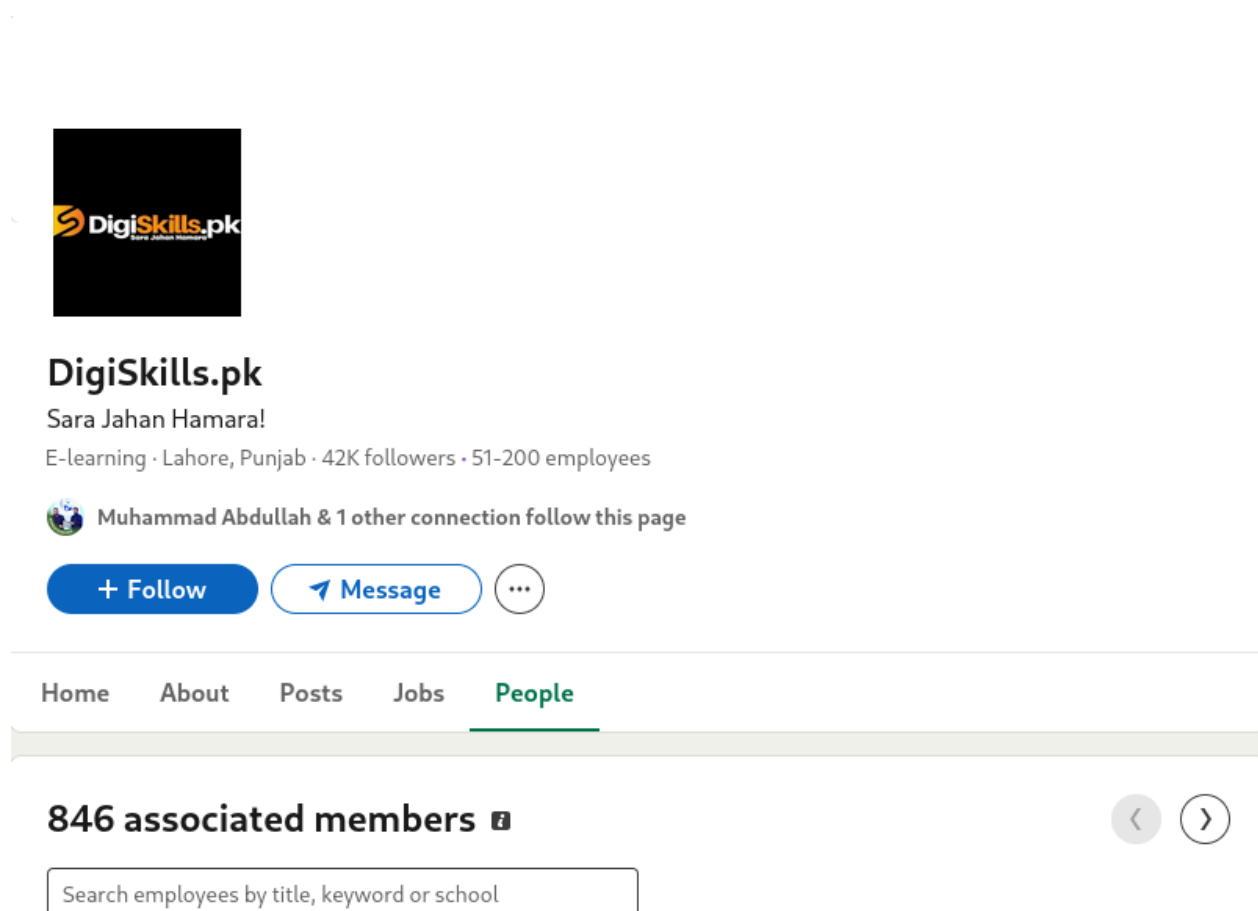


The screenshot displays a network analysis tool interface with the following sections:

- HTTP 4430 /TCP**
 - LAST OBSERVED: JUL 03, 2025 | 10:00 UTC
 - Details | Raw Data | JSON
- DETAILS**
 - URI: <https://58.27.207.132:4430/> [Go]
 - Status: 200 OK
 - Path: /
 - Body Hash: sha256:fa8bc340d599ce67b715caf65f52c75e4eef5da7f73e31e99c88251b96b0f
 - Response Body: <html> <head> <meta http-equiv="Pragma" content="no-cache"> <meta http-equiv="Cache-Control" content="no-cache" > <meta http-equiv="Cache-Control" content="must-revalidate" > <meta http-equiv="Expires" content="-1"> <meta http-equiv="Content-Type" content="text/html; charset=utf-8"> </head> <NOSCRIPT> Your browser does not support JavaScript (or it is... more </body>
- DETAILS**
 - URI: <https://58.27.207.132:4430/robots.txt> [Go]
 - Status: 200 OK
 - Path: /robots.txt
 - Body Hash: sha256:aa768c1c91a84875b1067e13ccad1a9ccbed13bffb0d223909cbb1c1dfa5b585
 - Response Body: User-agent: * Disallow: /
- TLS HANDSHAKE**
 - Version Selected: UNKNOWN
- CERTIFICATE**
 - Fingerprint: 012fab92104a5d514b903a7691e791e890452a499c6980cb590f89016464bb3
 - Subject DN: C=CN, ST= Guangdong, L= shenzhen, O= sangfor, OU= sslvpn, CN= sslvpn, emailAddress= ssl@sangfor.com
 - Issuer DN: C=CN, ST= Guangdong, L= shenzhen, O= sangfor, OU= sslvpn, CN= sslvpn, emailAddress= ssl@sangfor.com
 - Names: sslvpn

Company Profile: Digiskills.pk is an e-learning platform located in Lahore, Pakistan. It has 42,000 followers and is listed as having 51-200 employees. The CEO is Ali Zeenat.

Associated Individuals: The company's LinkedIn page shows 846 associated members. Some employees and their roles have been identified through public profiles, such as Aarish Fareed (Assistant Manager,Admin) and Tahir Siddique (Senior System/Network Administrator).



The screenshot displays the LinkedIn profile of DigiSkills.pk. The profile picture is a black square with the company's logo. The name 'DigiSkills.pk' is prominently displayed, followed by the tagline 'Sara Jahan Hamara!'. Below this, it states 'E-learning · Lahore, Punjab · 42K followers · 51-200 employees'. A notification indicates that 'Muhammad Abdullah & 1 other connection follow this page'. The navigation bar includes 'Home', 'About', 'Posts', 'Jobs', and 'People', with 'People' being the active tab. At the bottom, it shows '846 associated members' with left and right arrow icons, and a search bar with the placeholder text 'Search employees by title, keyword or school'.

DigiSkills.pk
Sara Jahan Hamara!
E-learning · Lahore, Punjab · 42K followers · 51-200 employees
Muhammad Abdullah & 1 other connection follow this page
+ Follow Message ...
Home About Posts Jobs **People**
846 associated members
Search employees by title, keyword or school

Employee Information: Mian Ali Hassan, is a Visual Designer and YouTuber who runs startups and is a course instructor for [Digiskills.pk](https://digiskills.pk).

[DigiSkills.pk](https://digiSkills.pk) is a free online training program in Pakistan that aims to equip young people with digital skills.

The public profiles of the individuals includes their names, job titles (“Ms Office Worker, Seo Specialist, Student, Copy writing”), locations (“Lahore, Peshawar, Daska, Gojra, Bhawalnagar”), and professional networks.

(Red Team)



Mian Ali Hassan • 3rd+
Visual Designer | YouTuber | Run Startups
3mo •

+ Follow ...

Behind the scenes at DigiSkills! 🎬

I was invited to record a course with them, and it's been an incredible experience working with such a professional team.

Grateful for the opportunity. DigiSkills.pk

If you're a freelancer, creator, or designer looking to grow — stay tuned. Something BIG is coming! 🚀📺📱

📍 Recorded with purpose, powered by experience.

[#DigiSkills](#) [#FreelancingPakistan](#) [#CourseInstructor](#) [#DigitalSkills](#) [#OnlineLearnin](#)



🔄❤️👤 24

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🔄 Repost

➦ Send

(Red Team)

**Said Arif** • 2nd
Ms Office worker
Peshawar
 Muhammad Abdullah Tariq is a mutual connection

**Ayesha Mustafa** • 2nd
SEO Specialist | Content writer | Canva Expert
Lahore
 Ayesha Aamir is a mutual connection

**Khizar Fayyaz** • 3rd+
 Boost Your Brand with Professional Designs | Freelance Canva Expert | Virtual Assistant |...
Daska

**Arfa Mubeen** • 3rd+
Student at Government College University, Faisalabad
Gojra

**Saqib Ali khan**  • 2nd
SEO (off page on page seo) Content creator Business development|Digital marketing|Email...
Lahore
 Faizan Ali and Ayesha Aamir are mutual connections

**Muhammad Yasir** • 2nd
Copywriting | Creative Writing | Blog Article |PowerPoint Presentation
Karak
 244 followers • Muhammad Abdullah Tariq is a mutual connection

**LinkedIn Member**
Amira Talha
Lahore District

**Uzma Siddique** • 2nd
I help businesses dominate search results and attract consistent, high-quality traffic with...
Bahawalnagar

Connect

Connect

Connect

Connect

Connect

Connect

Follow

Step 01 :

<Complete>

Step 02: Active Reconnaissance

TCP SYN Scan: Based on the Nmap scan, the target host at IP address **58.27.207.132**, identified as "mail.digiskills.pk", was found to be active and online. The scan did not identify any open ports but confirmed that the system is reachable.

Command Use: `sudo nmap -sn -PS 58.27.207.132`

```
(subhan@subhan)-[~]
$ sudo nmap -sn -PS 58.27.207.132

Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-04 12:33 PKT
Nmap scan report for mail.digiskills.pk (58.27.207.132)
Host is up (0.0071s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
```

Service Version Detection: A target at IP address **58.27.207.132** was scanned using Nmap to identify open services. The scanned revealed that the host is running a web server on port **80**. This service is identified as Microsoft **IIS httpd 10.0**. This information can be used for further reconnaissance and to identify potential vulnerabilities.

Command Use: `sudo nmap -sV -p 80 -Pn 58.27.207.132`

```
(subhan@subhan)-[~]
$ sudo nmap -sV -p 80 -Pn 58.27.207.132

Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-04 12:46 PKT
Nmap scan report for mail.digiskills.pk (58.27.207.132)
Host is up (0.0050s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http    Microsoft IIS httpd 10.0
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.09 seconds
```

OS Fingerprinting: Initial analysis reveals several open ports on the target server, “mail.digiskills.pk”, including ports for web traffic (80 and 443) and a potential chat service (6666 and 7777). This indicates that the server is communicating with the outside network through multiple services, which could be an entry point for an attacker. The scan also identified that one of the services is running “Apache httpd 2.4.18 (Ubuntu)” on port 8087, which is a known version and could have publicly documented vulnerabilities. An external scan identified a general-purpose router running a version of Linux or MikroTik RouterOS. While no exact operating system could be determined, this information provides a starting point for further reconnaissance to identify potential vulnerabilities.

Command Use: `sudo nmap -sS -sV -O --osscan-guess -T4 58.27.207.132`

```
(subhan@subhan)-[~]
$ sudo nmap -sS -sV -O --osscan-guess -T4 58.27.207.132
[sudo] password for subhan:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 16:52 PKT
Nmap scan report for mail.digiskills.pk (58.27.207.132)
Host is up (0.014s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
25/tcp    closed smtp
53/tcp    closed domain
80/tcp    open  tcpwrapped
110/tcp   closed pop3
443/tcp   open  ssl/tcpwrapped
6666/tcp  open  irc?
7777/tcp  open  cbt?
8087/tcp  open  http         Apache httpd 2.4.18 ((Ubuntu))
50389/tcp closed unknown

Device type: general purpose|router
Running (JUST GUESSING): Linux 2.6.X|3.X|4.X|5.X (86%), MikroTik RouterOS 7.X (85%)
OS CPE: cpe:/o:linux:linux_kernel:2.6 cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
Aggressive OS guesses: Linux 2.6.32 - 3.13 (86%), Linux 3.10 - 4.11 (86%), Linux 5.0 - 5.14 (86%), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3) (85%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 334.88 seconds
```


DNS Zone Transfer

01: The tool dnsrecon was used to scan the domain digiskills.pk and found that Domain Name System Security Extensions (**DNSSEC**) is not configured. The absence of DNSSEC makes the domain vulnerable to attacks like DNS Spoofing and Cache Poisoning, where an attacker could redirect traffic to a malicious site.

02: The scan also revealed that one of the DNS server, ns3.vu.edu.pk, is running BIND version "**9.18.34**". This version is known to have vulnerabilities, including Denial-of-Service (**Dos**) issued.

03: The scan exposed numerous **DNS** records, such as **MX** records for mail protection and **TXT** records with SPF and DMARC settings.

Command Use: `sudo dnsrecon -d digiskills.pk`

```
(subhan@subhan)-[~]
$ sudo dnsrecon -d digiskills.pk
[sudo] password for subhan:
[*] std: Performing General Enumeration against: digiskills.pk...
[-] DNSSEC is not configured for digiskills.pk
[*] SOA ns2.vu.edu.pk 58.27.207.130
[*] NS ns4.vu.edu.pk 111.68.104.220
[*] Bind Version for 111.68.104.220 "9.18.39"
[*] NS ns1.vu.edu.pk 111.68.103.45
[*] Bind Version for 111.68.103.45 "unknown"
[*] NS ns2.vu.edu.pk 58.27.207.130
[*] Bind Version for 58.27.207.130 "9.18.34"
[*] NS ns3.vu.edu.pk 58.27.231.130
[*] Bind Version for 58.27.231.130 "9.20.12"
[*] MX mail.digiskills.pk 58.27.207.132
[*] MX digiskills-pk.mail.protection.outlook.com 52.101.68.10
[*] MX digiskills-pk.mail.protection.outlook.com 52.101.68.0
[*] MX digiskills-pk.mail.protection.outlook.com 52.101.73.4
[*] MX digiskills-pk.mail.protection.outlook.com 52.101.68.18
[*] MX digiskills-pk.mail.protection.outlook.com 2a01:111:f403:ca09::9
[*] MX digiskills-pk.mail.protection.outlook.com 2a01:111:f403:ca04::1
[*] MX digiskills-pk.mail.protection.outlook.com 2a01:111:f403:ca09::b
[*] MX digiskills-pk.mail.protection.outlook.com 2a01:111:f403:ca09::a
[*] A digiskills.pk 58.27.207.132
[*] TXT digiskills.pk v=spf1 a mx ip4:172.16.104.199 ip4:58.27.207.132 ip4:172.16.104.205 ip4:111.68.104.205
com -all
[*] TXT _dmarc.digiskills.pk v=DMARC1; p=quarantine; rua=mailto:noreply@digiskills.pk; pct=100
[*] Enumerating SRV Records
[+] SRV _sipfederationtls._tcp.digiskills.pk sipfed.online.lync.com 52.123.173.219 5061
[+] 1 Records Found
```

Dig SOA Record

DNS SOA Record Details: A Start of Authority (SOA) record is a mandatory record for every DNS zone that contains administrative information about the domain.

Command Use: `dig SOA digiskills.pk`

Primary Name Server ([ns1.vu.edu.pk](#)): The primary DNS server for the zone. Secondary DNS servers get their updates from this server.

Responsible Person ([hostmaster.vu.edu.pk](#)): The email address of the person responsible for the domain's administration.

Serial Number (2025092402): This is a version number that secondary servers use to check if the zone has been updated.

Refresh (7200): The time in seconds (2 hours) that a secondary server waits before checking for changes to the SOA record on the primary server.

Retry (3600): The time, in seconds (1 hour), //

Expire (1209600): The time in seconds (14 days) that a secondary will continue to use the zone data if it cannot contact the primary server.

Minimum TTL (3600): The minimum Time-to-Live value in seconds (1 hour) for all records in the zone.

```
(subhan@subhan)-[~]
$ dig SOA digiskills.pk

; <<>> DiG 9.20.11-4+b1-Debian <<>> SOA digiskills.pk
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 29445
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 157de1fbfd94b9460100000068da7ff3307291fd9d026b60 (good)
;; QUESTION SECTION:
;digiskills.pk.                IN      SOA

;; ANSWER SECTION:
digiskills.pk.                41      IN      SOA      ns1.vu.edu.pk. hostmaster.vu.edu.pk. 2025092402 7200 3600 1209600 3600

;; Query time: 4 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Mon Sep 29 17:47:47 PKT 2025
;; MSG SIZE rcvd: 130
```

Dig NS Record

Command Use: **dig ns digiskills.pk**

NS (Name Server) Records: These records identify the DNS servers

ns3.vu.edu.pk

ns2.vu.edu.pk

ns1.vu.edu.pk

ns4.vu.edu.pk

A (Address) Records: These records map a domain or hostname to an IPv4 address.

ns1.vu.edu.pk is located at 111.68.103.45

ns2.vu.edu.pk is located at 58.27.207.130

ns3.vu.edu.pk is located at 58.27.231.130

ns4.vu.edu.pk is located at 111.68.104.220

```
(subhan@subhan)-[~]
$ dig ns digiskills.pk

; <<>> DiG 9.20.11-4+b1-Debian <<>> ns digiskills.pk
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 10015
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: d39bd58a54621d050100000068da7fd11ce34591ab37084e (good)
;; QUESTION SECTION:
;digiskills.pk.                IN      NS

;; ANSWER SECTION:
digiskills.pk.                54      IN      NS      ns3.vu.edu.pk.
digiskills.pk.                54      IN      NS      ns2.vu.edu.pk.
digiskills.pk.                54      IN      NS      ns1.vu.edu.pk.
digiskills.pk.                54      IN      NS      ns4.vu.edu.pk.

;; ADDITIONAL SECTION:
ns1.vu.edu.pk.                219     IN      A        111.68.103.45
ns2.vu.edu.pk.                219     IN      A        58.27.207.130
ns3.vu.edu.pk.                219     IN      A        58.27.231.130
ns4.vu.edu.pk.                219     IN      A        111.68.104.220

;; Query time: 4 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Mon Sep 29 17:47:13 PKT 2025
;; MSG SIZE rcvd: 215
```

Dig MX Record

Command Use: **dig MX digiskills.pk**

The **digiskills.pk** domain has two mail servers listed
digiskills-pk.mail.protection.outlook.com and **mail.digiskills.pk**.

DNS MX Record

Domain: **digiskills.pk**

Primary Mail Server: **digiskills-pk.mail.protection.outlook.com**

Primary Priority: **0**

Secondary Mail Server: **mail.digiskills.pk**

Secondary Priority: **50**

```
(subhan@subhan)-[~]
$ dig MX digiskills.pk

; <<>> DiG 9.20.11-4+b1-Debian <<>> MX digiskills.pk
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 47326
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 889a455c306cb4f70100000068da870c811eab545743c5a0 (good)
;; QUESTION SECTION:
;digiskills.pk.                IN      MX

;; ANSWER SECTION:
digiskills.pk.                360     IN      MX      0 digiskills-pk.mail.protection.outlook.com.
digiskills.pk.                360     IN      MX      50 mail.digiskills.pk.

;; Query time: 16 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Mon Sep 29 18:18:04 PKT 2025
;; MSG SIZE rcvd: 148
```

Dig TXT Record

Command Use: **dig txt digiskills.pk**

DNS TXT Record

Record Type: TXT (Text)

Domain: digiskills.pk

TTL(**Time-To-Live**): 300 seconds

Record content: "v=spf1 a mx ip4:172.16.104.199

ip4:58.27.207.132 ip4:172.16.104.205

pf.protection.outlook.com -all"

v=spf1:This indicates the record is a version 1 SPF record

a:This authorizes the **A** record (the main server IP) of the domain to send email

mx:This authorizes the **MX** record (mail servers) for the domain to send mail

ip4:172.16.104.199 ip4:58.27.207.132 ip4:172.16.104.205:These are specific IPv4 addresses that are authorized to send email for the domain

pf.protection.outlook.com:This authorize Microsoft Outlook's mail protection

-all:This is a "hard fail" qualifier

```
(subhan@subhan)-[~]
$ dig txt digiskills.pk

; <<>> DiG 9.20.11-4+b1-Debian <<>> txt digiskills.pk
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 37631
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: f0d0a7d8f07a22f80100000068da81b744d4f51a0431b592 (good)
;; QUESTION SECTION:
;digiskills.pk.                IN      TXT

;; ANSWER SECTION:
digiskills.pk.                300     IN      TXT     "v=spf1 a mx ip4:172.16.104.199 ip4:58.27.207.132 ip4:172.16.104.205
pf.protection.outlook.com -all"

;; Query time: 124 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Mon Sep 29 17:55:19 PKT 2025
;; MSG SIZE rcvd: 209
```

DNS Zone Transfer Via Nmap Script: A DNS zone transfer was successfully executed, which can expose all the subdomains and their corresponding IP addresses for a target network.

Command Use: `sudo nmap -p 53 --script dns-zone-transfer --script-args "dns-zone-transfer.domain=digiskills.pk" ns3.vu.edu.pk`

```
(subhan@subhan)-[~]
$ # Try zone-transfer via nmap script
sudo nmap -p 53 --script dns-zone-transfer --script-args "dns-zone-transfer.domain=digiskills.pk" ns3.vu.edu.pk

[sudo] password for subhan:
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 18:34 PKT
Nmap scan report for ns3.vu.edu.pk (58.27.231.130)
Host is up (0.011s latency).
rDNS record for 58.27.231.130: ns2.vu.edu.pk

PORT      STATE SERVICE
53/tcp    open  domain

Nmap done: 1 IP address (1 host up) scanned in 4.78 seconds
```

Banner Grabbing

Telnet: The server at IP address `58.27.207.132` is accessible on port `80`, but the initial request resulted in a `404 File or directory not found` error.

```
(subhan@subhan)-[~]
$ telnet 58.27.207.132 80
Trying 58.27.207.132 ...
Connected to 58.27.207.132.
Escape character is '^'.
GET /HTTP/1.0
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"/>
<title>404 - File or directory not found.</title>
<style type="text/css">
<!--
body{margin:0;font-size:.7em;font-family:Verdana, Arial, Helvetica, sans-serif;background:#EEEEEE;}
fieldset{padding:0 15px 10px 15px;}
h1{font-size:2.4em;margin:0;color:#FFF;}
h2{font-size:1.7em;margin:0;color:#CC0000;}
h3{font-size:1.2em;margin:10px 0 0 0;color:#000000;}
#header{width:96%;margin:0 0 0 0;padding:6px 2% 6px 2%;font-family:"trebuchet MS", Verdana, sans-serif;color:#FFF;
background-color:#555555;}
#content{margin:0 0 2%;position:relative;}
.content-container{background:#FFF;width:96%;margin-top:8px;padding:10px;position:relative;}
-->
</style>
</head>
<body>
<div id="header"><h1>Server Error</h1></div>
<div id="content">
<div class="content-container"><fieldset>
<h2>404 - File or directory not found.</h2>
<h3>The resource you are looking for might have been removed, had its name changed, or is temporarily unavailable.</h3>
</fieldset></div>
</div>
</body>
</html>
Connection closed by foreign host.
```

Ping Sweep of IP Range: The Scan was performed to discover if the host is online, but did not check for open ports or services.

Command Use: `sudo nmap -sn -PE 58.27.207.132`

```
(subhan@subhan)-[~]  
$ sudo nmap -sn -PE 58.27.207.132  
[sudo] password for subhan:  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-29 19:06 PKT  
Nmap scan report for mail.digiskills.pk (58.27.207.132)  
Host is up (0.0089s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.78 seconds
```

Step 02: **<Complete>**

Submit Task:

Email: `muhammadmoizuddinrafay@gmail.com`

(Red Team)